

Differentiate between Digital Computer and Quantum Computer.

1. Basic Unit of Information

- **Digital Computer (Classical):**
Uses a **bit** as the smallest unit of information.
 - A bit can only be in one of two states at a time: **0 or 1**.
 - **Quantum Computer:**
Uses a **qubit** (quantum bit).
 - Unlike bits, a qubit can exist in a **superposition** of 0 and 1 simultaneously.
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2. Processing Power

- **Digital Computer:**
Processes information sequentially or with limited parallelism
 - **Quantum Computer:**
Exploits **quantum parallelism**. Massive parallelism via superposition
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3. Fundamental Principles

- **Digital Computer:**
Works on **Boolean logic** (AND, OR, NOT) and deterministic algorithms.
 - **Quantum Computer:**
Works on Quantum gates (Hadamard, CNOT, etc.) and Probabilistic algorithms.
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4. Error and Noise

- **Digital Computer:**
Errors are rare due to stable hardware.
 - **Quantum Computer:**
Errors are high, requires error correction techniques.
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5. Applications

- **Digital Computer:**
General-purpose: word processing, browsing, gaming, databases, simulations, etc. Almost all daily life computing tasks.
 - **Quantum Computer:**
Specialized: Cryptography (breaking RSA using **Shor's Algorithm**) Database/search (faster search using **Grover's Algorithm**) Simulation of molecules and materials (useful in drug discovery, chemistry) Optimization problems (supply chains, finance, logistics)
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6. Current Status

- **Digital Computer:**
Fully developed and commercial
Mature, widely available, highly reliable. Billions of users worldwide.
- **Quantum Computer:**
Experimental, still under development
Still experimental, with small-scale prototypes (IBM, Google, Rigetti, etc.). Researchers are working to scale up qubits and reduce errors.

2. What is Shannon Entropy?

Shannon entropy was introduced by **Claude Shannon in 1948**. It quantifies the average amount of information produced by a random source. In other simple way we can say that it measures the **uncertainty or information content** in a random variable or message.

> **High Entropy:** When all outcomes are **unpredictable**, the uncertainty is maximized. For example, a fair coin toss has high entropy (1 bit).

> **Low Entropy:** When one outcome is much more likely than others, uncertainty is reduced. For example, a biased coin has low entropy.

> If an event is **certain** (probability = 1) then the Entropy is **0**.

system. It quantifies the average amount of information produced by a random source. The formula for the entropy $H(X)$ of a random variable X with outcomes $x_1, x_2, \dots, x_n$ and probabilities $p(x_1), p(x_2), \dots, p(x_n)$ is:

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

Where:

- $p(x_i)$ is the probability of each outcome.
- The logarithm is base 2, so the result is in **bits** (a unit of information).

Applications:

- **Data Compression:** Shannon entropy gives the *theoretical limit* of how much we can compress data without losing information.
- **Communication Systems:** Quantifies the amount of information to be transmitted.
- **Cryptography:** Measures the randomness of encryption keys or algorithms.
- **Machine Learning:** Used in decision tree algorithms to measure data "impurity."

3. What is a qubit and how does it differ from a classical bit?

A **qubit** (quantum bit) is the fundamental unit of information in **quantum computing**, and it differs significantly from a **classical bit** in how it stores and processes information.

> A classical bit can be in **only one of two definite states**, a qubit has two basic states, written as $|0\rangle$ and $|1\rangle$, But unlike a classical bit, a qubit can exist in a **superposition** of both states

For example, a qubit could be in a state like $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are **complex numbers**.

(Q.1 Answer)

Q.4 Write the general mathematical form of a qubit.

A **qubit** (quantum bit) is the basic unit of quantum information. The general mathematical form of a **qubit** is represented as a **superposition** of the two basis states $|0\rangle$ and $|1\rangle$. A qubit state can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where:

- $|\psi\rangle$ is the quantum state of the qubit.
- $|0\rangle$ and $|1\rangle$ are the computational basis states (often referred to as the **basis vectors**).
- α and β are **complex numbers** known as **probability amplitudes**.
- The normalization condition must hold:

$$|\alpha|^2 + |\beta|^2 = 1$$

This ensures that the total probability of measuring the qubit in either state is 100%

Column Vector Form

In vector notation, the qubit can be written as:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

So a general qubit state is:

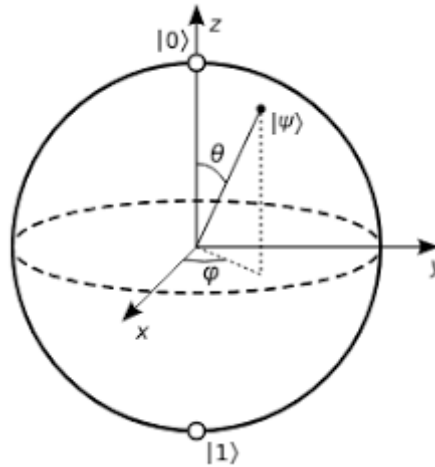
$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Q.5 What is the Bloch sphere and how does it represent a qubit?

The **Bloch sphere** is a geometric representation of the state of a **qubit**. It provides a way to visualize quantum states, specifically the superposition of the two computational basis states $|0\rangle$ and $|1\rangle$.

□ The Bloch sphere is a **unit sphere** in three-dimensional space, where each point on the sphere corresponds to a pure state of the qubit.

- The **north pole** of the sphere represents the state $|0\rangle$, and the **south pole** represents the state $|1\rangle$.
- Any other point on the surface of the sphere represents a **superposition** of $|0\rangle$ and $|1\rangle$, with the coefficients α and β corresponding to the angles θ and ϕ .



Mathematical Form:

We know a qubit can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with $|\alpha|^2 + |\beta|^2 = 1$.

This can be rewritten (using polar coordinates) as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

where:

- θ = angle from the **z-axis** (latitude)
- ϕ = angle around the **z-axis** (longitude)
- The factor $e^{i\phi}$ represents the **relative phase** between $|0\rangle$ and $|1\rangle$.

Q.6 Define a vector space in the context of quantum computing.

1. In general, a **vector space** is a collection of vectors that can be added together and multiplied by scalars (real or complex numbers), satisfying specific properties.

But in **quantum mechanics**, the states of a quantum system are represented by vectors in a **complex vector space (Hilbert space)**.

2. Qubit as a Vector:

- A **qubit** is a state vector in a **2-dimensional complex vector space**.

- The **basis vectors** of this space are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Any general qubit state is a **linear combination (superposition)** of these basis vectors:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with $\alpha, \beta \in \mathbb{C}$ and normalization $|\alpha|^2 + |\beta|^2 = 1$.

Thus, the qubit lives in a **2D complex vector space**.

3. Properties of Quantum Vector Spaces

1. Closure under addition:

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are qubit states, then $a|\psi_1\rangle + b|\psi_2\rangle$ (for complex numbers a, b) is also a valid state vector.

2. Inner Product:

Quantum vector spaces have an inner product, which allows us to compute probabilities and check orthogonality.

Example: $\langle 0|1\rangle = 0$, meaning $|0\rangle$ and $|1\rangle$ are orthogonal.

3. Dimension:

- 1 qubit $\rightarrow$ 2-dimensional vector space
- 2 qubits $\rightarrow 2^2 = 4$ -dimensional vector space
- n qubits $\rightarrow 2^n$ -dimensional vector space

This exponential growth is the source of quantum computing's power.

Q.7 What are the axioms of a vector space?

A **vector space** is a set of vectors (like quantum states) that satisfy certain properties under two operations: **vector addition** and **scalar multiplication**. These properties are formalized through **axioms** that the vectors and scalars must satisfy.

In quantum computing, the axioms of a vector space ensure that qubit states can be consistently added and scaled within a complex Hilbert space. This mathematical structure support superposition, unitary evolution, and measurement in quantum computing.

A. Axioms for Vector Addition

1. Closure under addition:

$$u + v \in V.$$

2. Commutativity:

$$u + v = v + u.$$

3. Associativity:

$$(u + v) + w = u + (v + w).$$

4. Existence of additive identity:

There is a zero vector $0 \in V$ such that $u + 0 = u$.

5. Existence of additive inverse:

For every $u \in V$, there exists $-u \in V$ such that $u + (-u) = 0$.

B. Axioms for Scalar Multiplication

6. Closure under scalar multiplication:

$$au \in V.$$

7. Distributivity of scalar multiplication over vector addition:

$$a(u + v) = au + av.$$

8. Distributivity of scalar multiplication over scalar addition:

$$(a + b)u = au + bu.$$

9. Associativity of scalar multiplication:

$$a(bu) = (ab)u.$$

10. Existence of multiplicative identity:

$1 \cdot u = u$, where 1 is the multiplicative identity in the field F .

Q.9 Describe linear combination of vectors?

A **linear combination** of vectors is a way of creating a new vector by **adding scalar multiples of other vectors**. In symbol :

$$v = a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

where:

- $v_1, v_2, \dots, v_n$ are vectors,
- $a_1, a_2, \dots, a_n$ are scalars (real or complex numbers),
- v is the resulting vector.

2. Linear Combination in Quantum Computing

In quantum computing, the concept of linear combination is used to describe **superposition of quantum states**.

- The computational basis for a qubit is $|0\rangle$ and $|1\rangle$.
- A general qubit state is a linear combination (or superposition) of these two basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha, \beta \in \mathbb{C}$.

- Example:

If $\alpha = \frac{1}{\sqrt{2}}$, $\beta = \frac{1}{\sqrt{2}}$, then:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

This means the qubit is in an equal superposition of $|0\rangle$ and $|1\rangle$.

Importance of Linear Combination

In quantum computing, any valid qubit state can be expressed as a **linear combination** of $|0\rangle$ and $|1\rangle$. It allows us to describe **all possible states** of a quantum system using a **basis**. For multiple qubits, we use linear combinations of basis states like $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ etc.

Q.11 Define the inner product between two quantum states.

In **quantum computing**, the **inner product** (also called the **scalar product**) between two quantum states is a crucial concept used to measure the similarity between the states. It is defined in the mathematical framework of **Hilbert space**

Inner Product Between Two Quantum States:

Let $|\psi\rangle$ and $|\phi\rangle$ be two quantum states (represented as vectors in a **Hilbert space**). The **inner product** between these two quantum states is denoted as:

$$\langle\psi|\phi\rangle$$

If we have two quantum states:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$$

their **inner product** is:

$$\langle\phi|\psi\rangle = \gamma^* \alpha + \delta^* \beta$$

3. Properties of Inner Product

1. Conjugate Symmetry:

$$\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$$

2. Linearity:

$$\langle \phi | (a|\psi_1\rangle + b|\psi_2\rangle) = a\langle \phi | \psi_1 \rangle + b\langle \phi | \psi_2 \rangle$$

3. Normalization:

Every valid quantum state is normalized:

$$\langle \psi | \psi \rangle = 1$$

4. Physical Meaning in Quantum Computing

- The **inner product** tells us how much one quantum state “overlaps” with another.
- If two states are **orthogonal** (like $|0\rangle$ and $|1\rangle$), then:

$$\langle 0 | 1 \rangle = 0$$

- If two states are **identical**, then:

$$\langle \psi | \psi \rangle = 1$$

- More generally, the **probability** of finding a system in state $|\phi\rangle$ when it is actually in state $|\psi\rangle$ is:

$$P = |\langle \phi | \psi \rangle|^2$$

Its squared magnitude, $|\langle \phi | \psi \rangle|^2$, gives the probability of transitioning from one state to the other.

Q.13 Define orthonormality in a vector space.

Orthonormality refers to a set of vectors that are both **orthogonal** and **normalized**. These properties are crucial in quantum mechanics as they help describe quantum states in a well-defined and consistent way

Orthonormality in a Vector Space:

Let a set of vectors in a **vector space**. These vectors are **orthonormal** if they satisfy the following two conditions:

1. Orthogonality

Two vectors $|u\rangle$ and $|v\rangle$ are called **orthogonal** if their inner product is zero:

$$\langle u | v \rangle = 0$$

👉 Meaning: they are completely independent, and share no overlap.

This means that the vectors are **non-interfering to each other**. Orthogonality ensures that the vectors are **linearly independent**.

2. Normalization:

A vector $|u\rangle$ is called **normalized** if its inner product with itself is 1:

$$\langle u|u\rangle = 1$$

👉 Meaning: the total probability of finding the state is 100%.

Example:

$$\langle 0|0\rangle = 1, \quad \langle 1|1\rangle = 1$$

This means the total probability of measuring a quantum state is 1 (i.e., the system is in a valid state).

Definition of Orthonormality:

Vectors are called **orthonormal** if they are both **orthogonal** and **normalized**:

$$\langle u|v\rangle = \begin{cases} 1 & \text{if } u = v \\ 0 & \text{if } u \neq v \end{cases}$$

Example of Orthonormal Vectors:

Consider two quantum states $|0\rangle$ and $|1\rangle$ (commonly used as the basis states for a qubit). These states are orthonormal because:

$$\langle 0|0\rangle = 1, \quad \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = 0, \quad \langle 1|0\rangle = 0$$

Thus, $\{|0\rangle, |1\rangle\}$ forms an **orthonormal** set of vectors.